


Feb 27, 2024.

Chomsky Normal Form (CNF)

A CFG is in **Chomsky Normal Form** if every rule in the grammar is of following form:

$$A \rightarrow BC$$

$$A \rightarrow a$$

where a is terminal symbol, $A, B, \& C$ are non terminal symbols.

Theorem: Any context-free language is generated by a context-free grammar in Chomsky-Normal Form.

Proof: Construction method to convert a grammar G into CNF.

Constructive proofs.

We are given a grammar G .

S

$$S_0 \rightarrow S$$

Remove ϵ -rules // ϵ -productions.

$$\boxed{A \rightarrow \epsilon}$$

$$\cancel{A \rightarrow a}, \cancel{A \rightarrow b}$$

$$X \rightarrow \alpha A \beta$$

$$X \rightarrow \alpha \beta$$

$$\cancel{A \rightarrow B}, \cancel{C \rightarrow D}$$



$$X \rightarrow ABC$$

$$\cancel{A \rightarrow D}$$

$$X \rightarrow DBC$$

$$X \rightarrow \overbrace{Y_1 Y_2 Y_3 \dots Y_n}$$

$$n \geq 2$$



$$X \rightarrow Y_1 Z_1$$

$$Z_1 \rightarrow Y_2 Z_2$$

$$Z_2 \rightarrow Y_3 Z_3$$

$$\vdots$$

$$\cancel{A \rightarrow BCDE}$$

$$\checkmark A \rightarrow BA' \quad (1)$$

$$\checkmark A' \rightarrow CA'' \quad (2)$$

$$\left[\begin{array}{l} A'' \rightarrow DA''' \\ A''' \rightarrow E \end{array} \right] \checkmark A'' \rightarrow DE \quad (3)$$

$$\left. \begin{array}{l} A \rightarrow aB \\ A \rightarrow XB \\ X \rightarrow a \end{array} \right\}$$

$$X \rightarrow A_1 A_2 A_3 A_4$$

$$X \rightarrow A_1 B_1$$

$$B_1 \rightarrow A_2 B_2$$

$$B_2 \rightarrow A_3 B_3 \quad B_2 \rightarrow A_3 A_4$$

$$B_3 \rightarrow A_4$$

input

$$S \rightarrow ASA | aB$$

$$A \rightarrow B | S$$

$$B \rightarrow b | \epsilon$$

$$S_0 \rightarrow S$$

$$S \rightarrow ASA | a \underline{B}$$

$$A \rightarrow \underline{B} | S$$

$$B \rightarrow b | \epsilon$$

$$X \underline{B \rightarrow \epsilon} X$$

$$S_0 \rightarrow S$$

$$S \rightarrow ASA | aB$$

$$A \rightarrow B | S | \underline{\epsilon}$$

$$B \rightarrow b$$

$$A \rightarrow \epsilon X$$

$$S_0 \rightarrow S$$

$$S \rightarrow \underline{ASA} | \underline{aB} | \underline{a} | \underline{SA} | \underline{AS} | \underline{S}$$

$$A \rightarrow \epsilon B | S_x$$

$$B \rightarrow b$$

$$S \rightarrow ASA \xleftarrow{X} A \rightarrow \epsilon X$$

$S \rightarrow SA$	✓
$S \rightarrow AS$	✓
$S \rightarrow S$	✓
$S \rightarrow \underline{ASA}$	

unit rules.

$$A \rightarrow B, A \rightarrow S$$

$$S_0 \rightarrow ASA | aB | a | SA | AS$$

$$S \rightarrow ASA | aB | a | SA | AS$$

$$A \rightarrow B^x | S | b$$

$$B \rightarrow b$$

→

$$S_0 \rightarrow ASA | aB | a | SA | AS$$
$$S \rightarrow ASA | aB | a | SA | AS$$
$$A \rightarrow S | b | ASA | aB | a | SA | AS$$
$$B \rightarrow b$$